Lecture 12 Pathological gait, part 2

SM2501/FSG3084



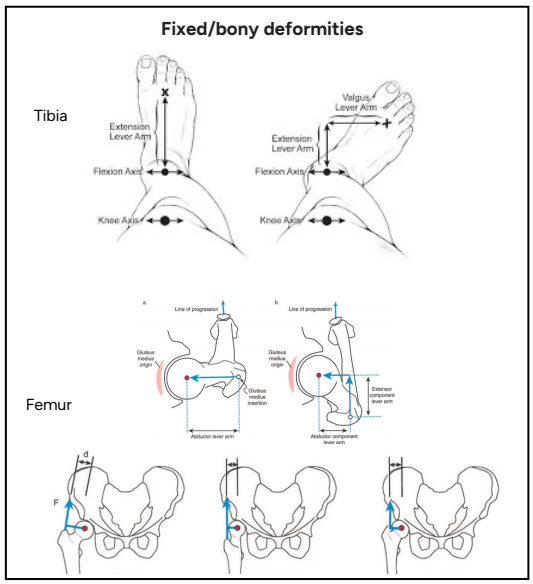
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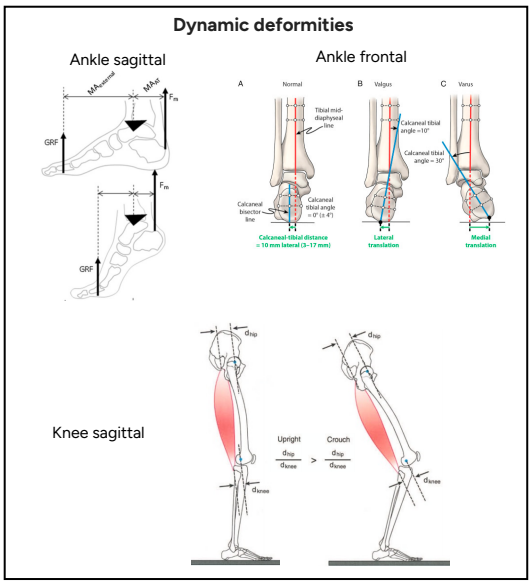
Why study gait in pathology?

- One of the most basic functions
  - Efficient, optimal energy conservation
  - "Patients strive to retain this capability even in the presence of severe impairment. As … pathology alter[s] mobility, … patients substitute wherever possible, yield when they must, and accept compensatory reactions…" (Jacquelin Perry 1992)

2

Lever arm dysfunction





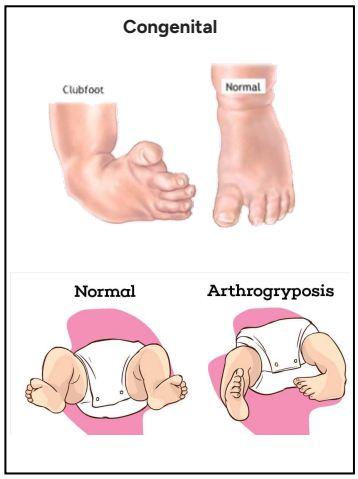
Paley et al 2002, Kalkman et al 2017

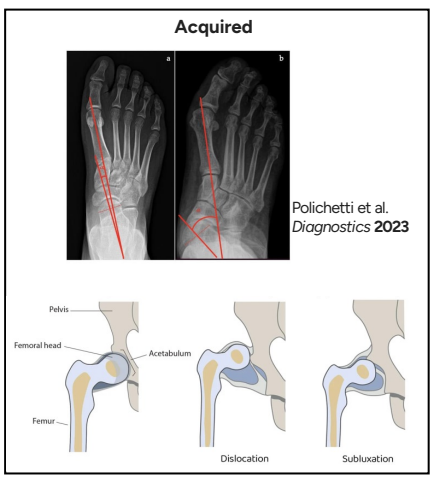
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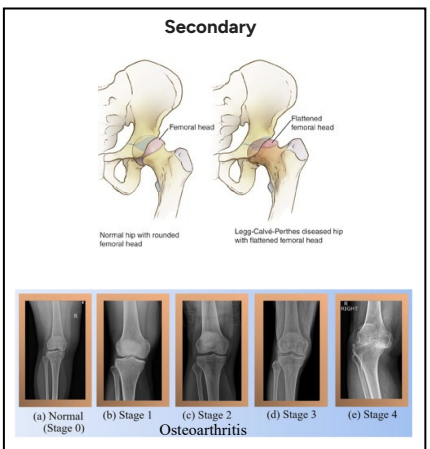
Joint deformity, bone and joint

Joint contracture: reduced joint ROM due to structural changes in fibrous connective tissue components of muscles, ligaments or joint capsule

- Abnormal joint and bone topology







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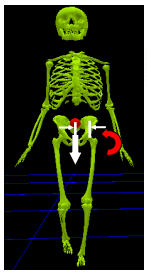
Pain

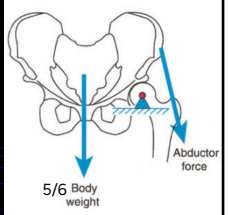


Often results in slow gait, avoidance of a side/joint, etc.

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Group exercise





Part 1: See the figures to the right of a person standing on 1 foot. Assume for this exercise that only the abductors are active at the hip, and can be modelled as 1 abductor muscle along this vector line of action.

- Rescale the Abductor muscle force vector length and approximate its magnitude in proportion BW. Make som reasonable assumption about relative moment arms.
- Draw a scaled vector of the hip joint contact force.

Part 2: Rearrange the body/segment positions so the abductor muscle force can (hypothetically) be reduced to 0. A stick figure of a whole body might help.

- Draw a new scaled vector of the hip joint contact force. Compare it to the joint contact force vector in the Part 1. Comments?

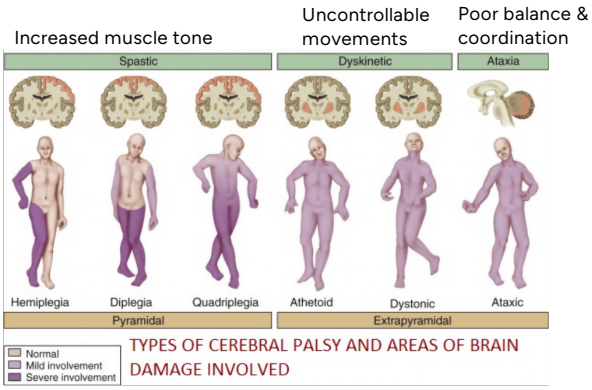
6

Gait Pathology Case Studies

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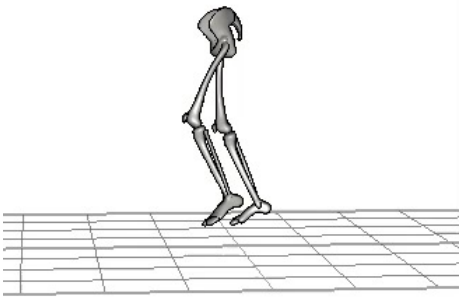
Cerebral palsy

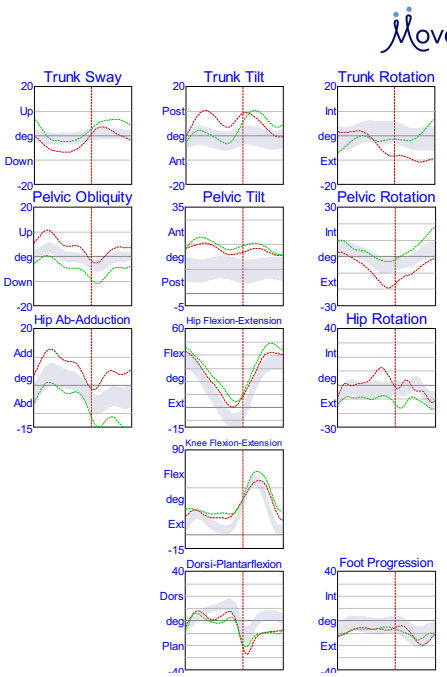
- Caused by an injury to the immature brain
- Usually occurs shortly before, during or shortly after birth
- Motor cortex highly vulnerable



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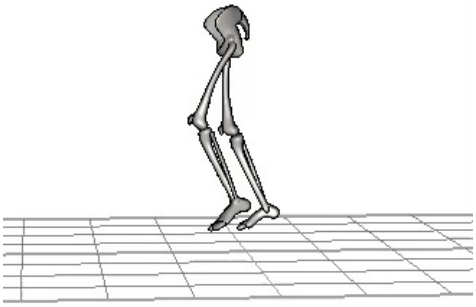
CP, mild severity
Gait kinematics

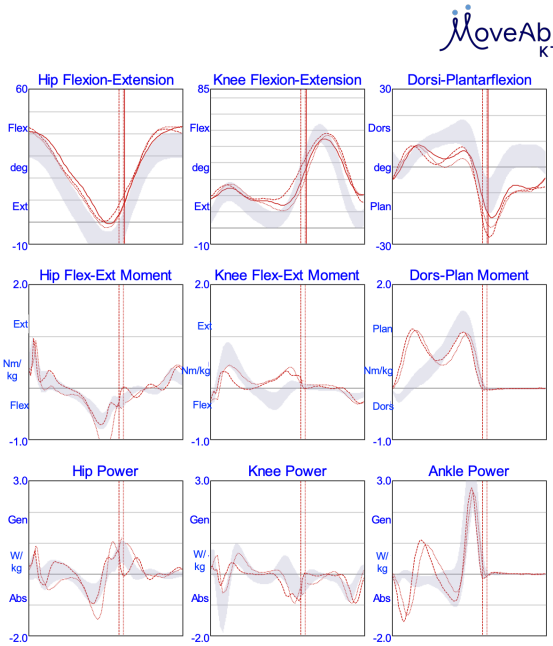




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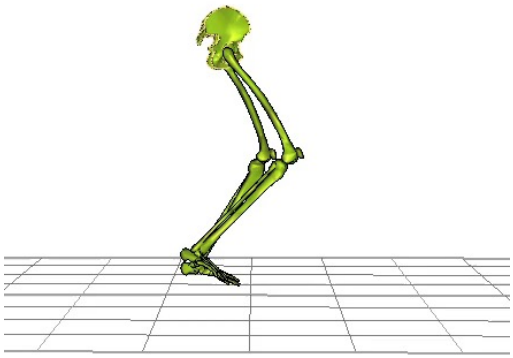
CP, mild severity
Gait kinetics



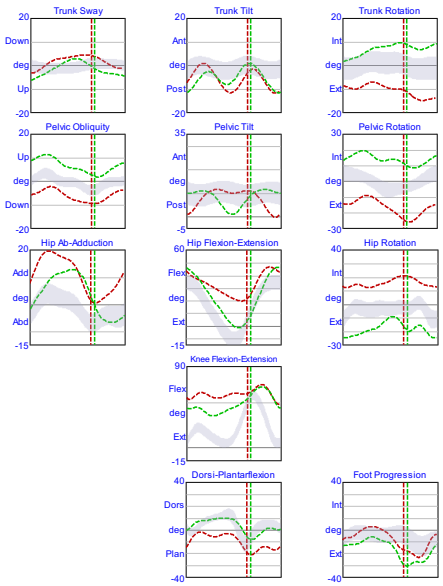


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CP, greater severity
Gait kinematics

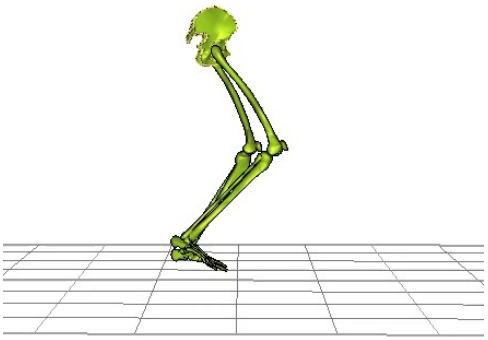


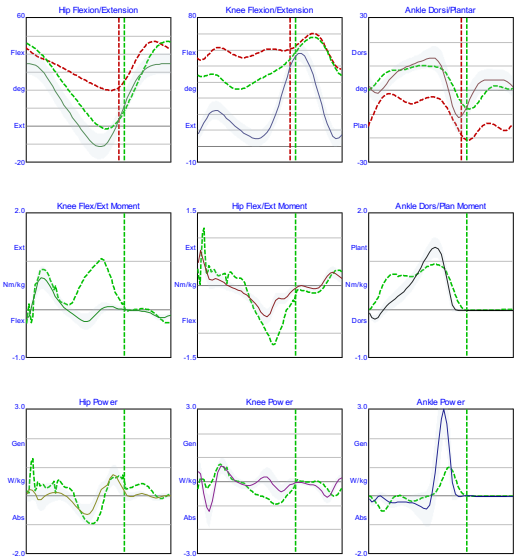
Gait kinematics



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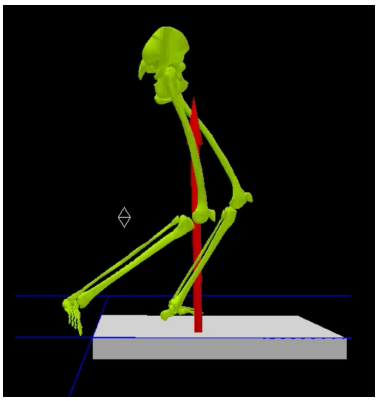
CP, greater severity
Gait kinematics

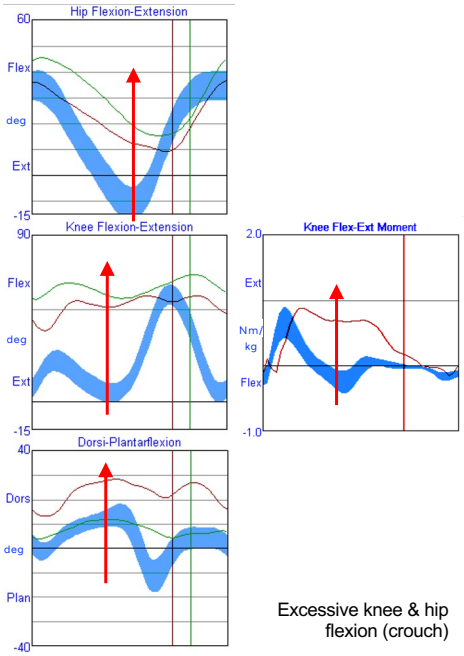




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CP diplegia, weak plantarflexors, short hamstrings

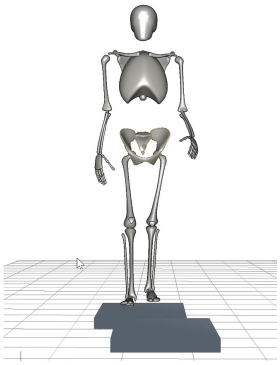




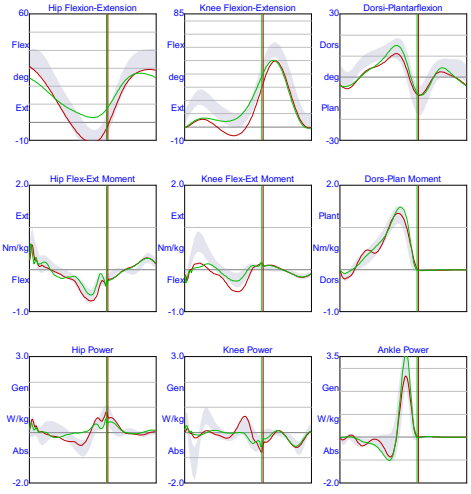
Excessive knee & hip flexion (crouch)

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Osteoarthritis

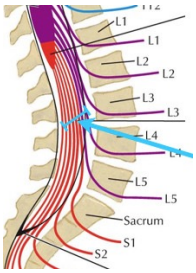


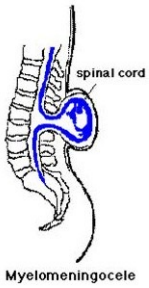
Right hip OA, pain

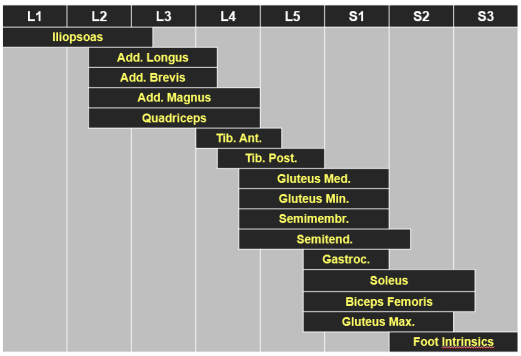


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Myelomeningocele







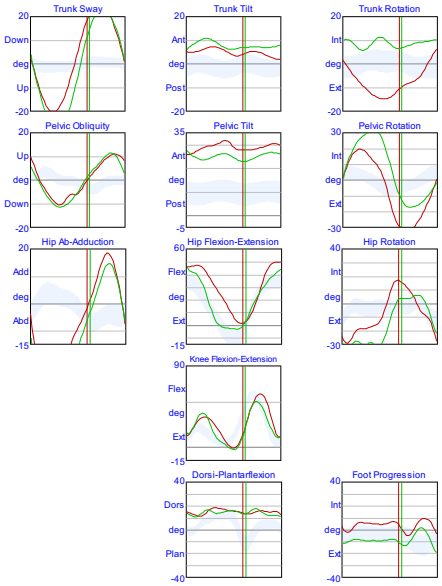
Innervation of lower limb muscles (Adapted from Sharrard 1964)

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Myelomeningocele

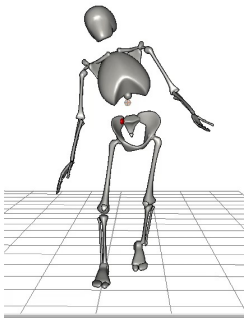
L4/L5: Weakness of hip abductors, hip extensors, plantarflexors and dorsiflexors

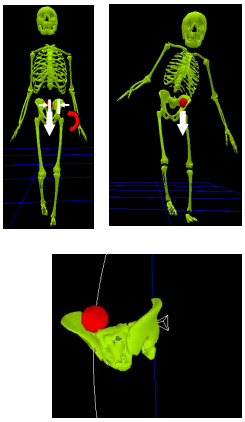


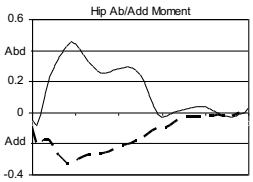


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Hip abductor-avoidance

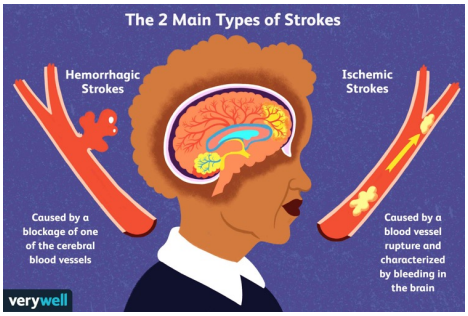






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Stroke



Damage to motor cells and CNS pathways.

Can result in disturbances of motor skills, balance, perception of equilibrium, processing of visual field.

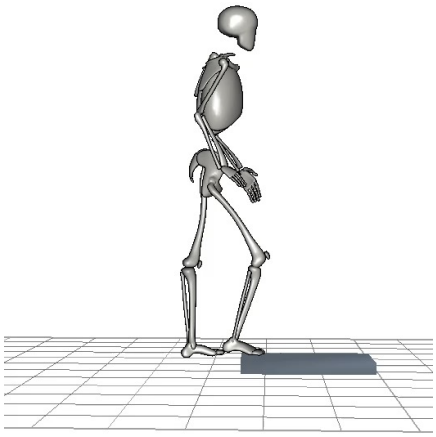
Can result in hemiparesis with changes in muscle tone (flaccid paresis and/or spasticity) on the involved side.

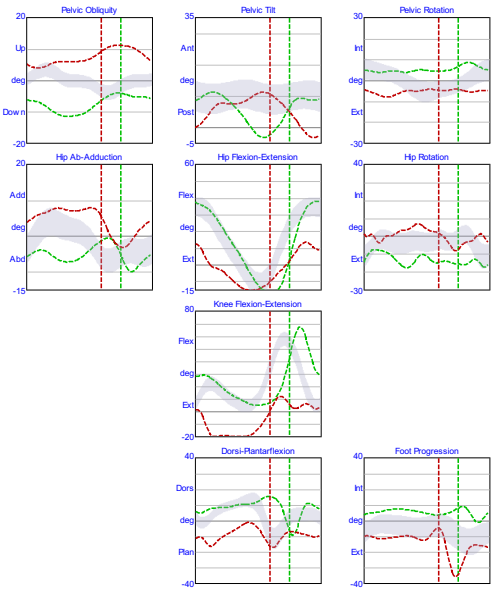
18

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Stroke

Left side affected
Stiff knee in swing

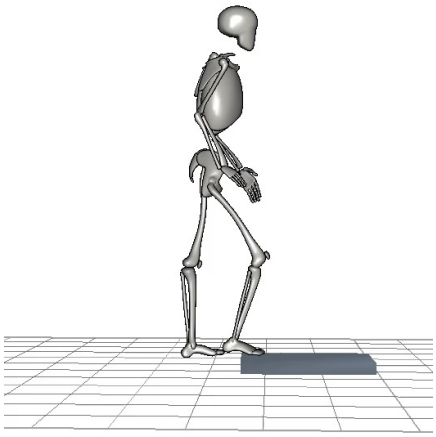


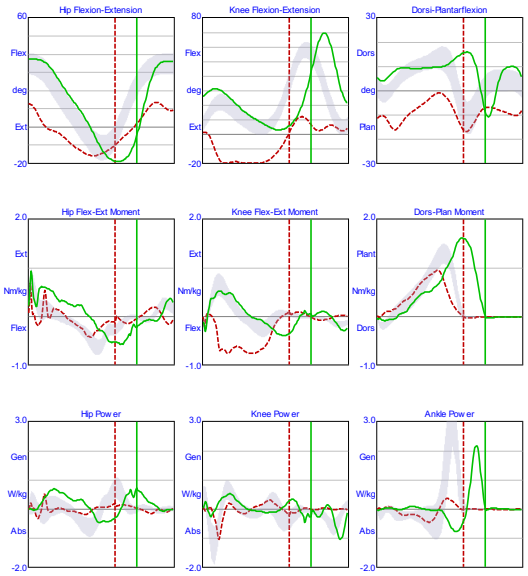


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Stroke

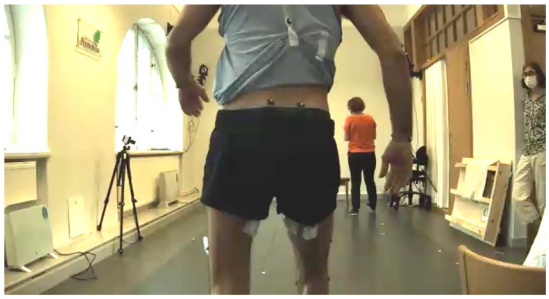
Left side affected
Stiff knee in swing





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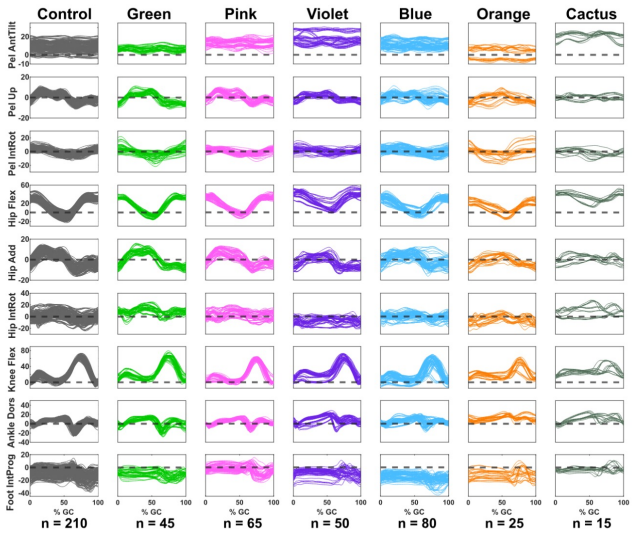
Incomplete spinal cord injury

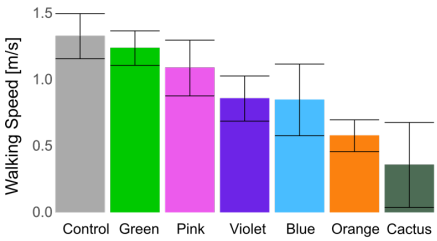


Motor deficits
Sensory deficits
Asymmetry

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Incomplete SCI



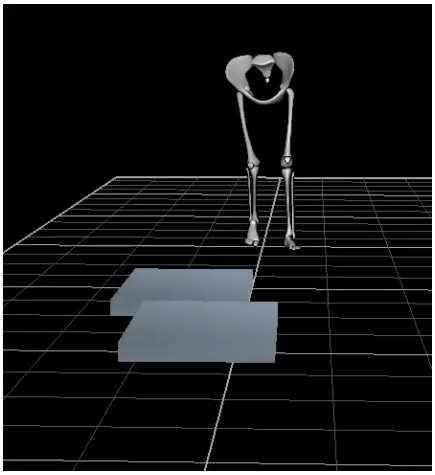


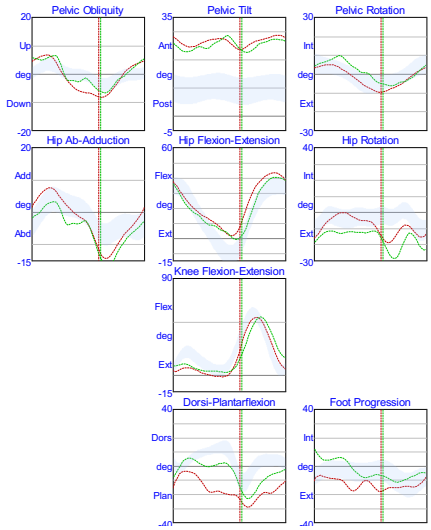
Truong et al., 2025

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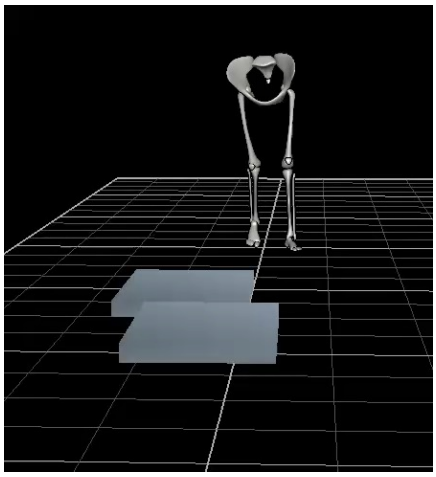
Duchenne Muscular Dystrophy

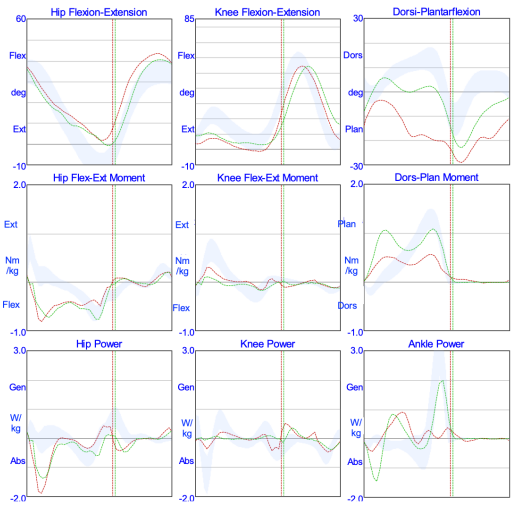
Muscle weakness lower extremities





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Challenges

Can we, using mechanical theory, give insight into optimal treatment?

Can tools be made more feasible for clinical professionals?

Can we simulate treatment (or lack thereof) in *short* and *long* term?

Can we design state-of-the-art movement assistive devices that address problem (only)?

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Mystery gaits

Gaits consist of:

Very fast walk
Very slow walk
Toe-walk
Stiff-knee walk
Crouch gait
Crouch gait on toes
Jog

In a group:

- Determine which number gait corresponds to the types of gait above.
- Identify characteristic patterns that help you determine this, in both kinematics and kinetics.
- Identify any other characteristic kinematic or kinetic pattern of interest in each walk.

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